Unit

11 Reactivity of metals

Unit Key Concepts

- Comparing the reactivity of metals
- The reactivity series
- Writing chemical and ionic equations
- Displacement reactions
- Prediction of metal reactions using the reactivity series

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

11.1 Comparing the reactivity of metals

In Unit 10, we mentioned that gold exists as a free element in the Earth's crust and iron as iron(III) oxide in haematite. Why does gold not exist as an oxide in its ore? Why does iron not exist as a free element in the Earth's crust?

The reason is that metals differ in their reactivity with other substances. To compare the reactivity of metals, we usually consider three aspects:

1 **The temperature at which the reaction starts** Many reactions start only when heat is supplied. The more reactive the metal is, the lower the temperature required.

2 **The rate of reaction** A more reactive metal reacts faster than a less reactive one.

3 **The amount of heat released during reaction** Generally, the more reactive the metal is, the more heat will be released during reaction.

We can compare the reactivity of metals based on their reactions with oxygen, water and dilute acids.

11.1

Comparing the reactivity of four unknown metals.

Video

- Reaction of potassium with air
- Reaction of sodium with air
- Reaction of magnesium with air

11.2 How do metals react with oxygen in the air?

Most metals react with oxygen in the air to produce oxides. The reactions usually require heat to start.

metal + oxygen ⟶ metallic oxide

N1 Examination questions often ask 'what would be observed' or 'observable change'. Give the observations, NOT the names of the products.

For example,

sodium + oxygen ⟶ sodium oxide

Table 11.1 lists the observations of heating some common metals N1 in the air. You can see that some metals react more readily than others. Metals react more rapidly in pure oxygen than in the air.

Iron(II) iron(III) oxide is a mixture of two oxides (iron(II) oxide and iron(III) oxide).

Table 11.1

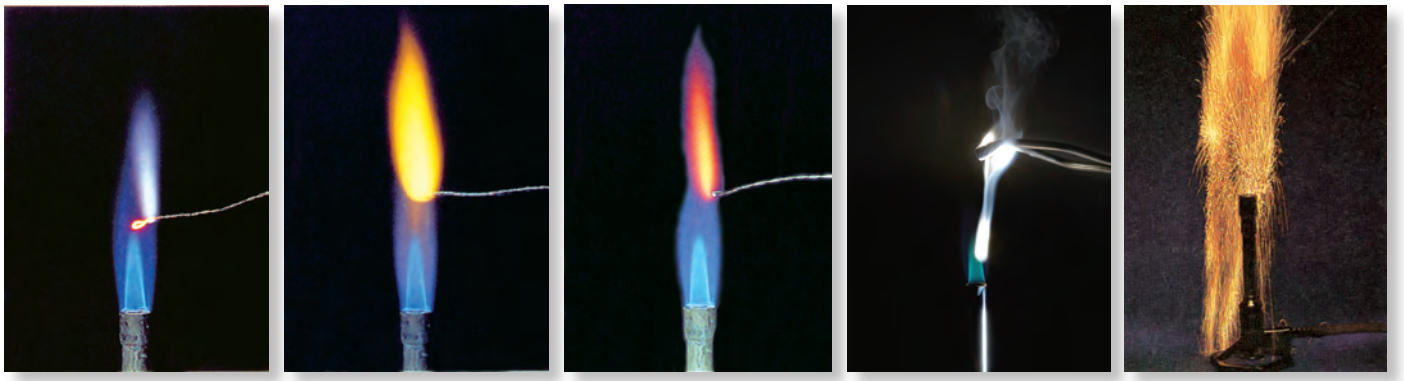
Observations of heating some metals in the air

Metal	Heat needed to start reaction	Observations	Oxide formed
Potassium (K)	gentle	• burns vigorously with a *lilac* flame • a white smoke is formed	potassium oxide (K_2O)
Sodium (Na)	gentle	• burns vigorously with a golden yellow flame • a white smoke is formed	sodium oxide (Na_2O)
Calcium (Ca)	strong	• burns quite vigorously with a brick-red flame • a white powder is formed N2	calcium oxide (CaO)
Magnesium (Mg)	strong	• burns with a very bright white light • a white powder is formed	magnesium oxide (MgO)
Aluminium (Al)	strong	• aluminium powder burns with white sparks • a white powder is formed	aluminium oxide (Al_2O_3)
Zinc (Zn)	strong	• a powder (yellow when hot but white when cold) is formed	zinc oxide (ZnO)
Iron (Fe)	strong	• iron powder burns with sparks • a black solid is formed	iron(II) iron(III) oxide (Fe_3O_4 or $FeO{\bullet}Fe_2O_3$)
Lead (Pb)	strong	• a powder (orange when hot but yellow when cold) is formed on the surface	lead(II) oxide (PbO)
Copper (Cu)	very strong	• a black powder is formed on the surface	copper(II) oxide (CuO)
Mercury (Hg)	very strong	• a red powder is formed on the surface	mercury(II) oxide (HgO)
Silver (Ag)	—	• no observable change	—
Platinum (Pt)	—	• no observable change	—
Gold (Au)	—	• no observable change	—

N2 Do NOT call the white powder formed a precipitate. The term 'precipitate' refers in the setting out of solid particles from a liquid (or from a gas).

lilac 淡紫色

Fig. 11.1 shows the appearance of flames produced when some metals burn in the air.



(a) Potassium burns with a lilac flame

(b) Sodium burns with a golden yellow flame

(c) Calcium burns with a brick-red flame

(d) Magnesium burns with a very bright white light

(e) Iron powder burns with sparks

Fig. 11.1 The appearance of flames produced when some metals burn in the air

CE 2009 Paper 1 Q2(a)(i)



Fig. 11.2 Sodium is so reactive that it is stored in paraffin oil

Many metals are dull in colour. This is because these metals reacted with oxygen in the air to form an oxide layer on the surface. When these metals are freshly cut or scratched, they appear shiny. Gold is the least reactive metal. It does not react with substances in the environment. Thus, it always appears shiny.

Potassium and sodium are very reactive. We store these metals in paraffin oil to prevent their reaction with oxygen in the air (Fig. 11.2).

11.3 How do metals react with cold water or steam? N3

N3 Examination questions often list some processes and ask which of the processes will produce a gas. For example, the following processes will produce hydrogen:
- passing steam over heated iron powder;
- adding zinc granules to dilute hydrochloric acid;
- electrolysis of acidified water.

Some metals react with cold water, but some only react with steam.

- Potassium, sodium and calcium react with cold water.
- Magnesium, aluminium, zinc and iron react with steam at elevated temperatures.
- Lead, copper, mercury, silver, platinum and gold do not react with cold water or steam at all.

Video
- Reaction of potassium with water
- Reaction of sodium with water
- Reaction of calcium with water

Reaction between cold water and potassium / sodium / calcium

Potassium, sodium and calcium react with cold water, producing metal hydroxides and hydrogen gas (Figs. 11.3–11.5).

metal + water ⟶ metal hydroxide + hydrogen

For example,

sodium + water ⟶ sodium hydroxide + hydrogen



Fig. 11.3 Potassium gives a lilac flame in cold water



Fig. 11.4 Sodium may give a golden yellow flame in cold water

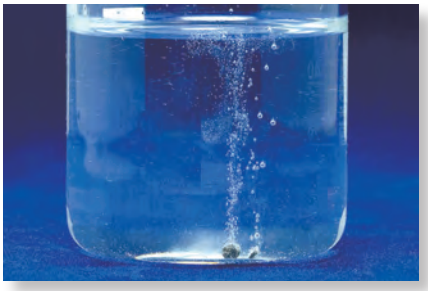


Fig. 11.5 Calcium reacts less vigorously with cold water than potassium and sodium do N4

N4 The evolution of the colourless gas may be quite slow at first. This is because the calcium metal is covered by a layer of calcium oxide. Reaction between the metal and water starts only when the oxide layer dissolves.

Table 11.2 lists the observations of reactions of potassium, sodium and calcium with cold water.

CE 2008 Paper 2 Q20
DSE 2014 Paper 1A Q18

Table 11.2

Observations of the reactions of some metals with cold water

Metal	Observations
Potassium (K)	• melts to form a silvery ball • fizzes furiously • moves rapidly on the water surface • the hydrogen produced in the reaction ignites and burns with a lilac flame◀
Sodium (Na)	• melts to form a silvery ball • fizzes quickly • moves rapidly on the water surface • sometimes the hydrogen produced in the reaction ignites and burns with a golden yellow flame◀
Calcium (Ca)	• calcium sinks in water • a steady stream of bubbles evolves

◀ Pure hydrogen burns with a blue flame. The flame appears lilac / golden yellow due to contamination of potassium / sodium compounds.

Example 11.1

Q Calcium reacts with cold water to give a gas. In a certain experiment, a student added some newly bought calcium granules into a beaker of cold water.

a) Write a word equation for the reaction between calcium and water.

b) Draw a labelled diagram of the set-up for carrying out the reaction of calcium with cold water in a beaker, with the collection of the gas.

c) The student added a few drops of universal indicator solution to the content of the beaker after the experiment.

State your expected observation. Explain your answer.

d) Suggest TWO potential hazards in performing the above experiment.

e) The student repeated the experiment with the same mass of an 'expired' sample of calcium granules that had been exposed in air for a long time.

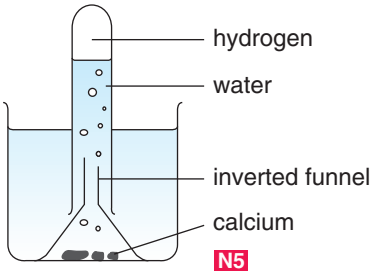
Suggest why less gas was collected in this second experiment when compared with the first experiment.

CE 2011 Paper 1 Q2

A

a) calcium + water ⟶ calcium hydroxide + hydrogen

b)



N5

c) The universal indicator solution changed from green to blue.

This is because the calcium hydroxide formed is alkaline.

d) Any two of the following:

- The hydrogen gas produced is explosive / flammable.
- Calcium is corrosive.
- Heat is released from the reaction.

e) The surfaces of the 'expired' sample of calcium granules had been oxidized by air. Calcium oxide was formed. Thus, less calcium reacted with the water to form hydrogen gas.

N5 Examination questions often ask about the reaction between calcium and water.
- Calcium sinks in water.
- An alkaline solution is formed in the beaker.

Students should be able to draw the experimental set-up for collecting the hydrogen gas evolved.

Reaction between steam and magnesium / aluminium / zinc / iron

Magnesium, aluminium, zinc and iron show little reaction with cold water. However, these metals react readily with steam. Metal oxides and hydrogen are produced during the reactions.

metal + steam ⟶ metal oxide + hydrogen

CE 2010 Paper 2 Q32

For example,

magnesium + steam ⟶ magnesium oxide + hydrogen

Fig. 11.6 shows the experimental set-up for the reaction between metal and steam.

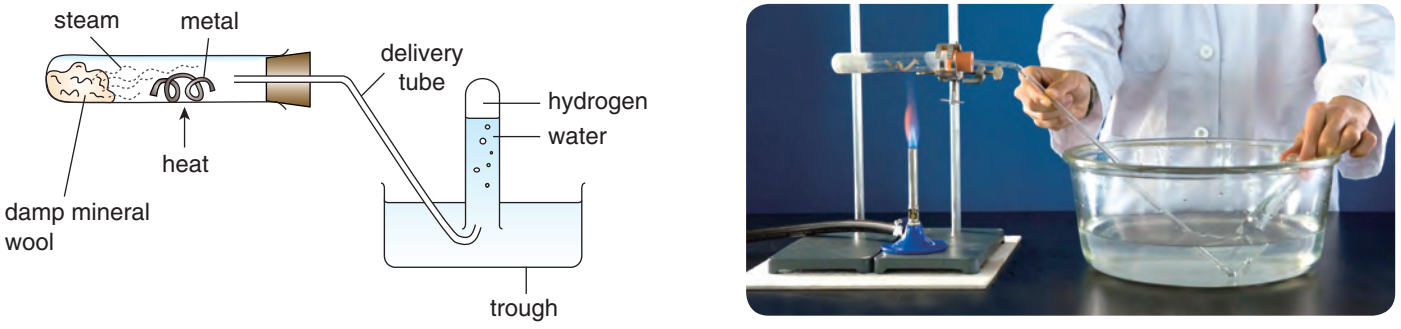


Fig. 11.6 Experimental set-up for the reaction between steam and magnesium / aluminium / zinc / iron

A strong heat is required for the metal to react with steam. There is no need to heat the damp mineral wool directly. Enough heat spreads back along the combustion tube to turn the water in the damp mineral wool into steam.

Table 11.3 lists the observations of reactions of some metals with steam.

Table 11.3

Observations of the reactions of some metals with steam

Metal	Observations
Magnesium (Mg)	• produces a very bright white light • a white powder remains
Aluminium (Al)	• reacts when the oxide layer is removed • a white powder is formed
Zinc (Zn)	• glows as steam is passed over, producing a yellow powder • the yellow powder becomes white when it is left to cool
Iron (Fe)	• heat until it glows red before steam is passed over • a black solid is formed

Aluminium is often covered with a thin oxide layer which prevents the reaction of the metal with steam. This oxide layer must be removed before any reaction can take place.

CE 2005 Paper 1 Q3(b)

11.4 How do metals react with dilute acids? N6

N6 Refer to page T29.

Reactive metals react with dilute acids (except dilute nitric acid[1]) to produce salts and hydrogen gas. For example, reactive metals react with dilute hydrochloric acid to produce metal chlorides and hydrogen gas. N7

[1] Metals react with dilute nitric acid to produce nitrogen monoxide. We will discuss this in Topic 5 Redox Reactions, Chemical Cells and Electrolysis.

N7 A counterfeit gold coin is made from an alloy of copper and zinc. We can use the following methods to show that the coin is not made of pure gold.
- Determining its density
- Treating it with dilute nitric acid
- Treating it with dilute hydrochloric acid

metal + dilute hydrochloric acid ⟶ metal chloride + hydrogen

For example,

magnesium + dilute hydrochloric acid ⟶ magnesium chloride + hydrogen

Reactive metals also react with dilute sulphuric acid to produce metal sulphates and hydrogen gas.

metal + dilute sulphuric acid ⟶ metal sulphate + hydrogen

For example,

iron + dilute sulphuric acid ⟶ iron(II) sulphate + hydrogen

CE 2010 Paper 2 Q32

Copper, mercury, silver, platinum and gold do not react with dilute hydrochloric acid and dilute sulphuric acid. Table 11.4 summarizes the observations of reactions of some metals with dilute hydrochloric acid.

Table 11.4

Observations of the reactions of some metals with dilute hydrochloric acid

Metal	Observations	Rate of reaction
Calcium (Ca)	• bubbles of gas are produced • the metal disappears • the test tube feels warm to the touch as heat is released	rate of production of bubbles increases ↑
Magnesium (Mg)		
Aluminium (Al)		
Zinc (Zn)		
Iron (Fe)		
Lead (Pb)[2]		

[2] Lead react with dilute hydrochloric acid to produce insoluble lead(II) chloride. The lead(II) chloride forms a protective layer on the surfaces of the metal. This layer prevents further reaction between the metal and the acid.

Video
- Reaction of calcium with dilute hydrochloric acid
- Reaction of magnesium with dilute hydrochloric acid

Fig. 11.7 shows how calcium, magnesium and iron react with dilute hydrochloric acid.



(a) Calcium reacts with dilute hydrochloric acid vigorously

(b) Magnesium reacts with dilute hydrochloric acid quickly

(c) Iron reacts with dilute hydrochloric acid slowly

Fig. 11.7 Reactions of calcium, magnesium and iron with dilute hydrochloric acid (from left to right)

11.5 The reactivity series of metals

We can use the reactivity of metals with air, cold water, steam and dilute acids to create a list of the metals in order of decreasing reactivity. This list is called the **reactivity series** of metals (Fig. 11.8). N8

N8 Refer to page T30.

Potassium and sodium react explosively with dilute acids. NEVER DO IT!

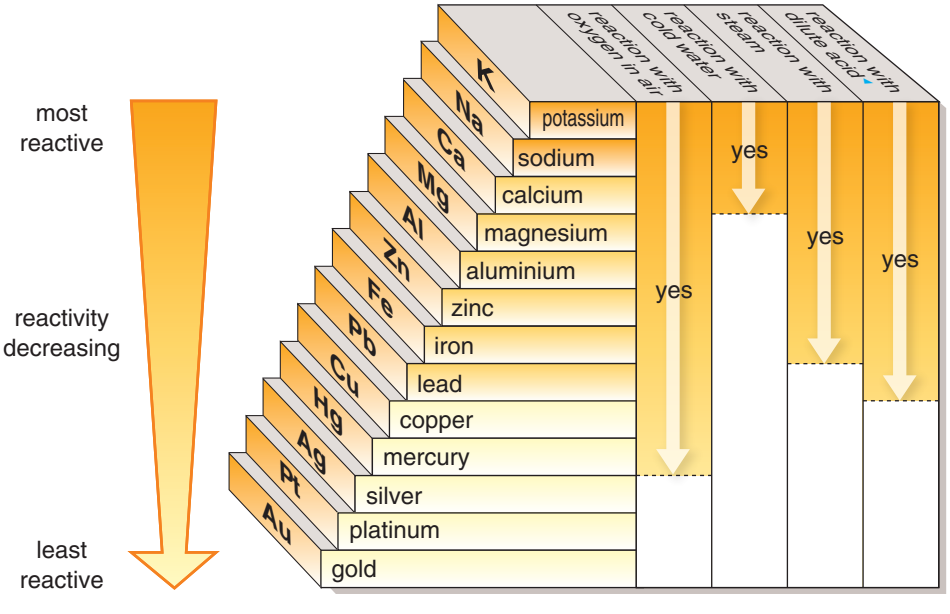


Fig. 11.8 Reactivity series of metals and the reactions of metals with air, cold water, steam and dilute acid

reactivity series 活性序

Practice 11.1

1 For each of the following cases, decide whether a reaction will take place.

a) Lead is added to cold water.
No reaction.

b) Magnesium is heated in oxygen.
Magnesium burns to give magnesium oxide.

c) Copper is added to dilute hydrochloric acid.
No reaction.

2 Titanium occurs as titanium(IV) oxide (*rutile* from beach sands). This oxide cannot be reduced to metal simply by heating with carbon. Titanium tarnishes slowly with oxygen but does not react with cold water.

a) Where would you place titanium in the reactivity series? Justify this positioning.
a) Above zinc but below calcium.
Harder to extract than zinc, but less reactive than calcium.

b) Historically, which metal, silver or titanium, would you expect to have become available earlier? Why?
Silver
It is easier to extract.

11.6 What is a balanced chemical equation?

When a burning splint is put into a test tube of hydrogen, the hydrogen burns with a 'pop' sound. The hydrogen reacts with oxygen in the air to form water. In this reaction, hydrogen and oxygen are the **reactants**, whereas water is the **product**.

The word equation for the reaction is:

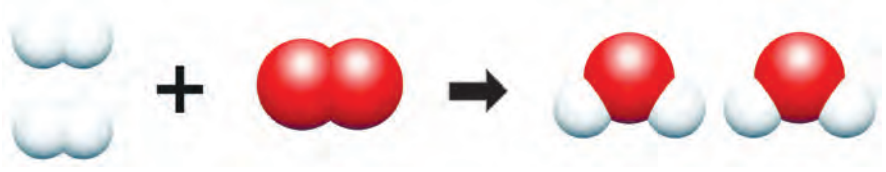
hydrogen + oxygen ⟶ water

In a word equation, we write the reactant(s) on the left while the product(s) is / are written on the right. A plus (+) sign on the left means 'to react with'. A single arrow (⟶) means 'is / are changed to'.

However, a word equation does not give the relative amount of each substance involved in the reaction. Hence chemists use **chemical equations** to represent reactions. In a chemical equation, we use a combination of chemical symbols and formulae to represent a reaction.

rutile 金紅石 reactant 反應物 product 生成物 chemical equation 化學方程式

In the above reaction, for every two hydrogen molecules that react, one oxygen molecule is needed and two water molecules are formed.



2 hydrogen molecules (reactant) | 1 oxygen molecule (reactant) | 2 water molecules (product)

The following chemical equation can represent the reaction between hydrogen and oxygen:

$$2H_2 + O_2 \longrightarrow 2H_2O$$

We must understand the difference between the big numbers (or coefficients) written in front of the chemical formulae, such as the '2' in $2H_2$, and the smaller subscripts such as the '2' in H_2O.

The big number '2' in $2H_2$ tells us that there are 2 hydrogen molecules. The '2' in H_2O tells us that there are 2 hydrogen atoms in one water molecule. Look at the number of atoms on each side of the equation:

$2H_2$ + O_2 ⟶ $2H_2O$

4 hydrogen atoms 2 oxygen atoms		4 hydrogen atoms 2 oxygen atoms

The numbers of hydrogen and oxygen atoms are the same on both sides of the equation. This is because atoms do not disappear during a reaction. They just rearrange themselves to form new substances.

When the numbers of all atoms are the same on both sides, a chemical equation is **balanced**.

balanced 平衡的

Adding state symbols to chemical equations

The states of reactants may affect the results of a reaction. Therefore we should write down the states of substances in a chemical equation. We use (s), (l), (g) and (aq) to represent solid, liquid, gas and aqueous solution respectively.

The complete balanced chemical equation for the reaction between hydrogen and oxygen is:

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$$

The single arrow ($\longrightarrow$) in a chemical equation shows that the reaction goes one way only. We can use a double arrow ($\rightleftharpoons$) for a reversible reaction. A **reversible reaction** is one that can go in either direction, depending on the conditions.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

What information can you obtain from the following balanced chemical equations? N9

1 $Mg(s) + H_2O(g) \longrightarrow MgO(s) + H_2(g)$

2 $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Refer to page T31.

N9 A chemical equation does not tell us the conditions and the rate of a reaction. For example,
$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l)$
The above equation does not tell us the conditions for the reaction to occur. In fact, a mixture of hydrogen and oxygen would remain unreacted at room temperature. The mixture will explode if a flame is applied.

11.7 How to write balanced chemical equations

If we know the reactant(s) and product(s) of a reaction, we can find their chemical formulae and then write the balanced chemical equation.

Rules for writing balanced chemical equations

To write a balanced chemical equation for a reaction, we should follow the steps outlined below. Take the reaction between aluminium and oxygen as an example. Aluminium burns in oxygen to form aluminium oxide.

reversible reaction 可逆反應

Step	Example
1 Write down the word equation.	aluminium + oxygen ⟶ aluminium oxide
2 Write down the chemical formulae of the reactant(s) and product(s).	$Al + O_2 \longrightarrow Al_2O_3$
3 Balance the chemical equation so that the number of each type of atom on one side is equal to that on the other side. a) To balance the aluminium atom, put the coefficient '2' before Al.	$2Al + O_2 \longrightarrow Al_2O_3$
b) To balance the oxygen atom, put the coefficient '$\frac{3}{2}$' before O_2. Now the chemical equation is balanced.	$2Al + \frac{3}{2}O_2 \longrightarrow Al_2O_3$ **(Balanced)**
c) To get a whole number coefficient for each substance, multiply all coefficients by 2.	$4Al + 3O_2 \longrightarrow 2Al_2O_3$ **(Balanced)**
4 Write down the state symbol after each substance.	$4Al(s) + 3O_2(g) \longrightarrow 2Al_2O_3(s)$ **(Balanced)** N10

The chemical formula of oxygen is O_2 as there are two atoms in one oxygen molecule.

N10 Refer to page T31.

Here are some more guidelines for balancing chemical equations:

1 Consider a polyatomic ion (e.g. SO_4^{2-}, NO_3^- and HCO_3^-) as a single unit if it appears on both sides of the equation. For example, if there are two sulphate ions (SO_4^{2-}) on one side, we shall need two on the other side.

2 We can only find out the product(s) of a reaction by doing experiments. We cannot determine the product(s) of a reaction simply by writing a chemical equation.

Example 11.2

Q Write a balanced chemical equation for the reaction between zinc and dilute hydrochloric acid to form zinc chloride solution and hydrogen gas.

A

1 Write down the word equation.	zinc + hydrochloric acid ⟶ zinc chloride + hydrogen
2 Write down the chemical formulae of the reactants and products.	$Zn + HCl \longrightarrow ZnCl_2 + H_2$
3 To balance the chlorine atom, put the coefficient '2' before HCl. Now the chemical equation is balanced.	$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$ **(Balanced)**
4 Write down the state symbol after each substance.	$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$ **(Balanced)**

Example 11.3

Q Write a balanced chemical equation for the reaction of iron(III) oxide with carbon monoxide to form iron and carbon dioxide in a blast furnace.

A

1 Write down the word equation.	iron(III) oxide + carbon monoxide $\longrightarrow$ iron + carbon dioxide
2 Write down the chemical formulae of the reactants and products.	$Fe_2O_3 + CO \longrightarrow Fe + CO_2$
3 a) To balance the iron atom, put the coefficient '2' before Fe.	$Fe_2O_3 + CO \longrightarrow 2Fe + CO_2$
b) To balance the oxygen atom, put the coefficient '3' before CO and CO_2. Now the chemical equation is balanced.	$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$ **(Balanced)**
4 Write down the state symbol after each substance.	$Fe_2O_3(s) + 3CO(g) \longrightarrow 2Fe(s) + 3CO_2(g)$ **(Balanced)**

Example 11.4

Q Write a balanced chemical equation for the reaction between silver nitrate solution and copper to form silver and copper(II) nitrate solution.

A

1 Write down the word equation.	silver nitrate + copper $\longrightarrow$ silver + copper(II) nitrate
2 Write down the chemical formulae of the reactants and products.	$AgNO_3 + Cu \longrightarrow Ag + Cu(NO_3)_2$
3 a) To balance the polyatomic ion NO_3^-, put the coefficient '2' before $AgNO_3$.	$2AgNO_3 + Cu \longrightarrow Ag + Cu(NO_3)_2$
b) To balance the silver atom, put the coefficient '2' before Ag.	$2AgNO_3 + Cu \longrightarrow 2Ag + Cu(NO_3)_2$ **(Balanced)**
4 Write down the state symbol after each substance.	$2AgNO_3(aq) + Cu(s) \longrightarrow 2Ag(s) + Cu(NO_3)_2(aq)$ **(Balanced)**

Practice 11.2

1 Write a balanced chemical equation, including state symbols, for each of the following reactions.

a) Solid calcium carbonate reacts with dilute hydrochloric acid to form an aqueous solution of calcium chloride, water and carbon dioxide.
$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

b) Sodium reacts with water to form an aqueous solution of sodium hydroxide and hydrogen.
$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$

c) Solid sodium hydrogencarbonate decomposes upon heating to form solid sodium carbonate, water and carbon dioxide.
$2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + H_2O(l) + CO_2(g)$

2 Balance the following chemical equations.

a) $CuO(s) + HCl(aq) \longrightarrow CuCl_2(aq) + H_2O(l)$
$CuO(s) + 2HCl(aq) \longrightarrow CuCl_2(aq) + H_2O(l)$

b) $Pb(s) + AgNO_3(aq) \longrightarrow Pb(NO_3)_2(aq) + Ag(s)$
$Pb(s) + 2AgNO_3(aq) \longrightarrow Pb(NO_3)_2(aq) + 2Ag(s)$

c) $FeS(s) + O_2(g) \longrightarrow Fe_2O_3(s) + SO_2(g)$
$4FeS(s) + 7O_2(g) \longrightarrow 2Fe_2O_3(s) + 4SO_2(g)$

11.8 What determines the reactivity of a metal?

We have learnt that different metals have different reactivities. What causes such differences?

A common feature of all the reactions of metals with oxygen, cold water and dilute acids is that atoms of the metals lose electrons to become cations.

For example,

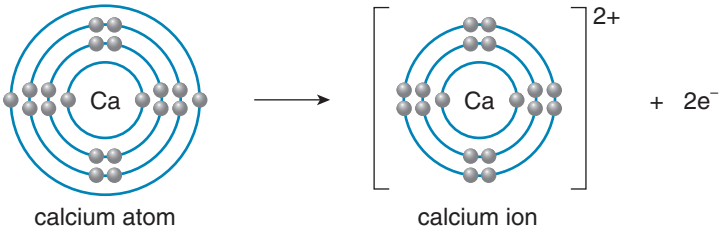
- magnesium reacts with oxygen to form magnesium oxide, which contains magnesium ions and oxide ions;
- calcium reacts with cold water to form calcium hydroxide, which contains calcium ions and hydroxide ions;
- zinc reacts with dilute hydrochloric acid to form zinc chloride, which contains zinc ions and chloride ions.

The relative ease with which a metal loses its outermost shell electrons is a major factor affecting its reactivity.

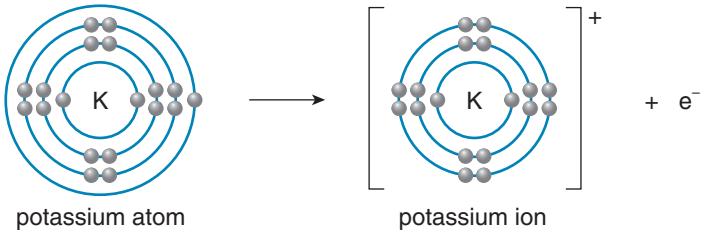
✓ Atoms of more reactive metals lose outermost shell electrons to form cations more readily.

Comparing the reactivity of potassium, sodium, calcium and aluminium

A calcium atom has two outermost shell electrons, while both potassium and sodium atoms have one outermost shell electron. During a reaction, a calcium atom needs to lose two electrons while both potassium and sodium atoms need to lose only one electron (Fig. 11.9). The process of electron losing requires energy. Losing one electron is easier than losing two electrons. Hence potassium and sodium are more reactive than calcium.



a) A calcium atom loses two electrons during a reaction



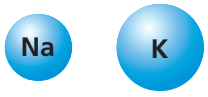
b) A potassium atom loses one electron during a reaction

Fig. 11.9 Calcium and potassium atoms lose electrons during reactions

Similarly, an aluminium atom will lose three outermost shell electrons during a reaction. This is more difficult than losing two electrons as more energy is required. Thus, calcium is more reactive than aluminium.

Comparing the reactivity of potassium and sodium

A potassium atom has four occupied electron shells. A sodium atom has three occupied electron shells. Hence a potassium atom is larger than a sodium atom. The diagram below shows the relative size of sodium and potassium atoms.



The size of the atom of a metal also affects how easily the atom loses electrons. A potassium atom is larger than a sodium atom. The outermost shell electron of a potassium atom is further away from the nucleus than that of a sodium atom. There is less attraction between the outermost shell electron and the nucleus in a potassium atom. A potassium atom loses its outermost shell electron more readily than a sodium atom. Hence potassium is more reactive than sodium.

Trends in the reactivity of metals in the periodic table

The readiness of metals to lose electrons increases as we move down a group and decreases as we go across a period in the periodic table (Fig. 11.10).

The reactivity of the Group I metals increases in the order Li < Na < K while that of Group II metals increases in the order Be < Mg < Ca.

Across the third period, reactivity decreases in the order Na > Mg > Al.

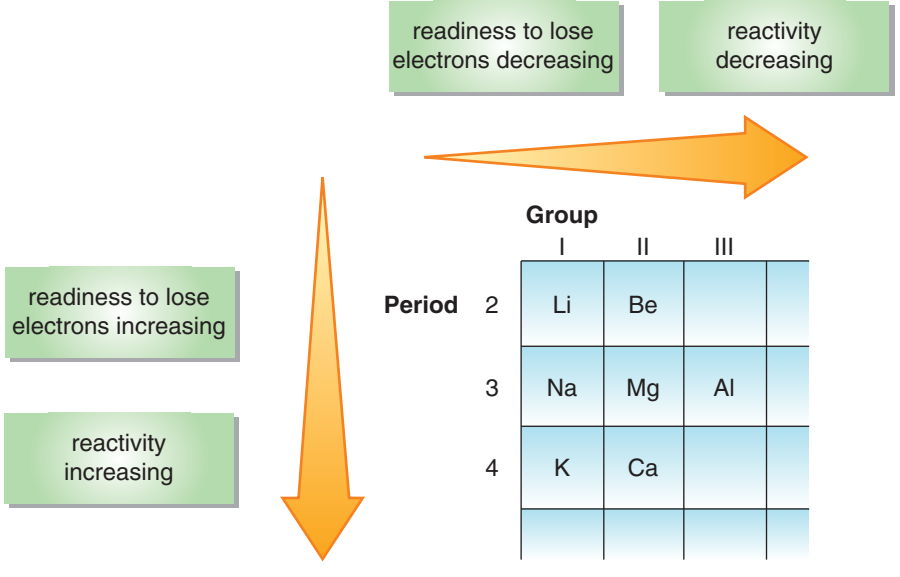


Fig. 11.10 Trends in the reactivity of metals in the periodic table

11.2

Comparing the reactivity of metals by their displacement reactions.

Video
Comparing the reactivity of metals using displacement reactions

Animation
Displacement reaction

11.9 Displacement reactions

When we dip a piece of copper into a colourless silver nitrate solution, the copper metal slowly becomes coated with a grey deposit of silver. The colour of the solution becomes blue gradually (Fig. 11.11). This is because the solution contains copper(II) ions. However, when we place a piece of silver into a copper(II) nitrate solution, no reaction takes place.

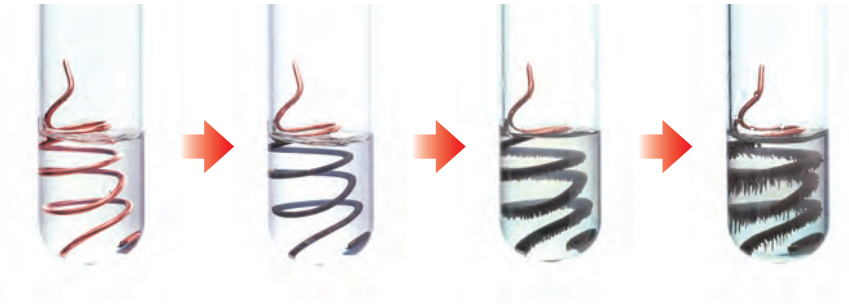


Fig. 11.11 Copper displaces silver from silver nitrate solution. The solution is blue in colour because it contains copper(II) ions

N11 Do NOT confuse 'reduction with carbon' with 'displacement reaction'.

A reaction in which one element displaces another element from a compound is called a **displacement reaction**. N11

CE 2006 Paper 2 Q13
CE 2007 Paper 2 Q7
CE 2008 Paper 2 Q34

✓ A metal will always displace a less reactive metal from a solution of the compound of the less reactive metal. N12

N12 When excess magnesium ribbons are added to iron(II) sulphate solution, the solution gradually changes from pale green to colourless. A displacement reaction occurs.

In the above reaction, copper displaces silver from silver nitrate solution to form copper(II) nitrate solution and silver deposits.

$$Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

Copper is more reactive than silver. A copper atom loses electrons more easily than a silver atom. When copper reacts with silver nitrate solution, two electrons from each copper atom are transferred to two silver ions. Thus, the copper atom changes into a copper(II) ion and the two silver ions change into silver atoms. The solution becomes copper(II) nitrate solution.

displacement reaction 置換反應

Similarly, zinc displaces copper from copper(II) sulphate solution to form zinc sulphate solution and copper deposits (Fig. 11.12).

$$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$$



Fig. 11.12 Reaction between zinc and copper(II) sulphate solution

Example 11.5

Q X, Y and Z are three different metals. The table below shows the results of two experiments carried out using the metals.

Experiment	Metal		
	X	Y	Z
Adding the metal to cold water	a colourless gas was given off	no observable change	no observable change
Adding the metal to copper(II) sulphate solution	a colourless gas was given off and a reddish brown deposit formed	no observable change	a reddish brown deposit formed

a) Name the colourless gas given off when X was added to cold water.

b) Name the type of reaction that occurred when Z was added to copper(II) sulphate solution.

c) Arrange the three metals in order of increasing reactivity. Explain your answer.

d) Predict whether X or Y was discovered earlier. Explain your answer.

e) Explain why a colourless gas was given off when X was added to copper(II) sulphate solution.

Continued on next page

A1 a) Hydrogen

b) Displacement reaction

c) Reactivity: Y < Z < X

X was the most reactive as it gave a colourless gas with cold water but the other two did not. Z was more reactive than Y as it could displace copper from copper(II) sulphate solution but Y could not.

d) Y was discovered earlier. It is less reactive than X and thus easier to extract.

e) X is a reactive metal. It reacted with the water in the copper(II) sulphate solution to give hydrogen gas.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Problem Solving

Suppose you are given samples of four metals (W, X, Y, Z) and a dilute solution of each of the metal nitrates ($W(NO_3)_2(aq)$, $X(NO_3)_2(aq)$, $Y(NO_3)_2(aq)$, $Z(NO_3)_2(aq)$).

Suggest how you would compare the reactivity of the four metals using displacement reactions.

Refer to page T32.

11.10 Ionic equations

We can represent the displacement reaction between copper and silver nitrate solution by the following balanced chemical equation:

$$Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

The copper and the silver are metals consisting of atoms. However, the silver nitrate and the copper(II) nitrate are metal compounds and so are ionic. The silver nitrate solution contains $Ag^+(aq)$ and $NO_3^-(aq)$ ions while the copper nitrate solution contains $Cu^{2+}(aq)$ and $NO_3^-(aq)$ ions.

We can rewrite the equation showing the ions:

$$Cu(s) + 2Ag^+(aq) + 2NO_3^-(aq) \longrightarrow Cu^{2+}(aq) + 2NO_3^-(aq) + 2Ag(s)$$

The nitrate ions do not take part in the reaction. We can delete them from the equation. The equation for the reaction thus becomes:

$$\underset{\text{(net charge = +2)}}{Cu(s) + 2Ag^+(aq)} \longrightarrow \underset{\text{(net charge = +2)}}{Cu^{2+}(aq) + 2Ag(s)}$$

Notice that the net charge on the left-hand side of the equation is the same as that on the right-hand side.

An equation like this, which involves only ions formed or changed during a reaction, is called an **ionic equation**. The main advantage of an ionic equation is that it shows clearly which ions are taking part in a reaction.

Example 11.6

Q Write an ionic equation for the reaction between iron and dilute hydrochloric acid.

A

1 Write down the word equation.	iron + dilute hydrochloric acid ⟶ iron(II) chloride + hydrogen
2 Write down the chemical formulae of the reactants and products.	$Fe + HCl \longrightarrow FeCl_2 + H_2$
3 Balance the chemical equation and write down the state symbol after each substance.	$Fe(s) + 2HCl(aq) \longrightarrow FeCl_2(aq) + H_2(g)$ **(Balanced)**
4 HCl and $FeCl_2$ exist as mobile ions in solution. Rewrite them to show the ions involved. (Fe is a solid and H_2 is a gas. Do not change their chemical formulae.)	$Fe(s) + 2H^+(aq) + \cancel{2Cl^-(aq)} \longrightarrow Fe^{2+}(aq) + \cancel{2Cl^-(aq)} + H_2(g)$ **(Balanced)**
5 Delete ions that are not taking part in the reaction to obtain the ionic equation.	$\underset{\text{(net charge = +2)}}{Fe(s) + 2H^+(aq)} \longrightarrow \underset{\text{(net charge = +2)}}{Fe^{2+}(aq) + H_2(g)}$ **(Balanced)**

Practice 11.3

1 Write an ionic equation for each of the following reactions:

a) calcium is added to dilute hydrochloric acid; and
$Ca(s) + 2H^+(aq) \longrightarrow Ca^{2+}(aq) + H_2(g)$

b) magnesium is added to silver nitrate solution.
$Mg(s) + 2Ag^+(aq) \longrightarrow Mg^{2+}(aq) + 2Ag(s)$

2 Manganese (Mn) lies between aluminium and zinc in the reactivity series. Solutions of manganese(II) salts are almost colourless (or pale pink).
Refer to page T32.

a) Use the reactivity series to predict whether manganese will react with copper(II) sulphate solution. If it will react, state your expected observations and write an ionic equation for the reaction.

b) Explain why you would expect manganese to react with steam. Name the products of the reaction and write a chemical equation for the reaction.

ionic equation 離子方程式

11.11 Reactivity of metals and ease of extraction

How a metal is extracted depends to a large extent on its position in the reactivity series.

✓ The lower the position of a metal in the reactivity series, the more easily it can be extracted from its ores.

Gold is at the bottom of the series. It can be extracted by physical methods. Metals near to the bottom of the series, such as copper, mercury and silver, can be extracted from sulphide ores simply by heating the sulphides in air.

For example,

mercury(II) sulphide + oxygen $\xrightarrow{\text{heat}}$ mercury + sulphur dioxide

$HgS(s) + O_2(g) \xrightarrow{\text{heat}} Hg(l) + SO_2(g)$

For more reactive metals such as zinc and lead, their sulphides are first roasted in air to form the oxides. Their oxides can then be reduced by heating with carbon.

For example,

zinc sulphide + oxygen $\xrightarrow{\text{roast in air}}$ zinc oxide + sulphur dioxide

$2ZnS(s) + 3O_2(g) \xrightarrow{\text{roast in air}} 2ZnO(s) + 2SO_2(g)$

zinc oxide + carbon $\xrightarrow{\text{heat}}$ zinc + carbon dioxide

$2ZnO(s) + C(s) \xrightarrow{\text{heat}} 2Zn(s) + CO_2(g)$

Iron is obtained from its oxide ore by heating with carbon monoxide.

Iron(III) oxide + carbon monoxide $\xrightarrow{\text{heat}}$ iron + carbon dioxide

$Fe_2O_3(s) + 3CO(g) \xrightarrow{\text{heat}} 2Fe(s) + 3CO_2(g)$

For the most reactive metals, potassium to aluminium in the reactivity series, electrolysis of molten compounds must be used.

Fig. 11.13 summarizes the extraction methods for metals in the reactivity series.

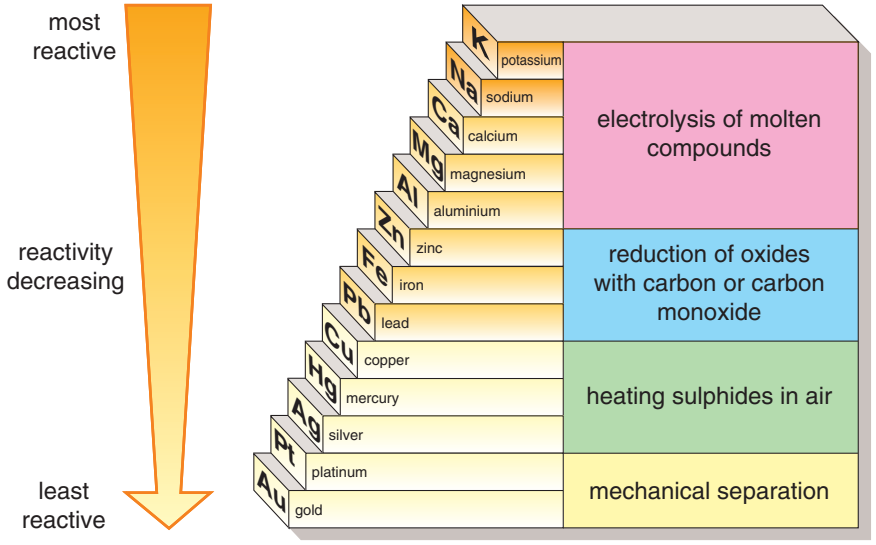


Fig. 11.13 Reactivity of some metals and extraction of the metal compounds

11.12 Prediction of metal reactions using the reactivity series

Reactivity series and reduction of metal oxides

11.3
Extracting silver by heating.

Metals react with oxygen in the air to form metal oxides. Metal oxides can be reduced to metals again.

CE 2006 Paper 2 Q13
DSE 2014 Paper 1B Q4

For example, silver oxide is reduced to silver by heating alone N13 (Fig. 11.14). Lead(II) oxide is more difficult to reduce. We can obtain lead by heating lead(II) oxide with carbon (Fig. 11.15).

Video
Extracting silver from silver oxide

$$\text{silver oxide} \xrightarrow{\text{heat}} \text{silver} + \text{oxygen}$$

$$2Ag_2O(s) \xrightarrow{\text{heat}} 4Ag(s) + O_2(g)$$

CE 2005 Paper 1 Q2(a)(i)

$$\text{lead(II) oxide} + \text{carbon} \xrightarrow{\text{heat}} \text{lead} + \text{carbon dioxide}$$

$$2PbO(s) + C(s) \xrightarrow{\text{heat}} 2Pb(s) + CO_2(g)$$

N13 Mercury(II) oxide is also reduced to mercury by heating alone.



Fig. 11.14 Silver oxide is reduced to silver by heating alone



Fig. 11.15 Reduction of lead(II) oxide with carbon

DSE 2014 Paper 1B Q4

✓ Compounds of a less reactive metal are more unstable. Thus, it is easier to reduce the oxide of the metal. N14

N14 In general, the more reactive the metal, the greater the energy required to extract it from its ores.

Metal	Main metallic compound in ore	Energy required to produce 1 kg of metal (kJ)
Al	Al_2O_3	31 100
Fe	Fe_2O_3	7 370
Cu	CuO	2 470

The position of silver is lower than that of lead in the reactivity series. Thus, it is easier to reduce silver oxide than lead(II) oxide.

Competition for oxygen between metals

CE 2007 Paper 1 Q6(b)

Besides reducing a metal oxide with carbon, we can also reduce the oxide of a less reactive metal by heating it with a more reactive metal.

For example, when a mixture of iron(III) oxide powder and aluminium powder is ignited, a very vigorous reaction occurs. This reaction is known as the **thermit reaction** (Fig. 11.16).

thermit reaction 鋁熱反應

$$\text{iron(III) oxide} + \text{aluminium} \xrightarrow{\text{ignite}} \text{iron} + \text{aluminium oxide}$$

$$Fe_2O_3(s) + 2Al(s) \xrightarrow{\text{ignite}} 2Fe(s) + Al_2O_3(s)$$

Aluminium is more reactive than iron. It can remove oxygen from iron(III) oxide. N15

N15 Aluminium in NOT used as the reducing agent in iron extraction because aluminium is more expensive than iron.

On the other hand, iron cannot remove oxygen from aluminium oxide.

DSE 2013 Paper 1B Q7(a) & (b)



Fig. 11.16 Rails are welded together using molten iron produced by the thermit reaction

A large amount of heat is released in the reaction and the iron formed is in molten state.

Practice 11.4

1 X, Y and Z are three different metals. When these metals are placed separately into iron(II) sulphate solution, a layer of iron is formed only on Y. When each of the oxides of these metals is heated strongly, only the oxide of X gives a metallic lustre. Arrange the three metals in increasing order of reactivity. Explain your answer.

X < Z < Y

Only Y can displace iron from iron(II) sulphate solution. Thus, Y is the most reactive.

Only the oxide of X decomposes when heated strongly. Thus, X is the least reactive.

2 In an experiment, copper(II) oxide is reduced by heating with magnesium.

a) Write a chemical equation for the reaction involved.

$CuO(s) + Mg(s) \longrightarrow Cu(s) + MgO(s)$

b) Suggest whether silver can be used to replace magnesium in the reduction of copper(II) oxide.

No. The reactivity of silver is lower than that of copper. Thus, silver cannot reduce copper(II) oxide.

Have you mastered?

Key terms

metal chloride 金屬氯化物 36 metal sulphate 金屬硫酸鹽 36 reactivity series 活性序 37

reactant 反應物 38 product 生成物 38 balanced chemical equation 平衡化學方程式 38

state symbols 物態符號 40 reversible reaction 可逆反應 40 displacement reaction 置換反應 46

ionic equation 離子方程式 49 thermit reaction 鋁熱反應 52

Checklist

After studying this unit, you should be able to

- ☐ describe and compare the reactions of some common metals with oxygen/air, cold water and dilute acids;
- ☐ write the word equations for the reactions of metals with oxygen/air, cold water and dilute acids;
- ☐ construct a metal reactivity series with reference to their reactions, if any, with oxygen/air, cold water and dilute acids;
- ☐ write balanced chemical equations to describe various reactions;
- ☐ use the state symbols (s), (l), (g) and (aq) to write chemical equations;
- ☐ relate the reactivity of metals to the tendency of metals to form positive ions;
- ☐ describe and explain the displacement reactions involving various metals and metallic compounds in aqueous solutions;
- ☐ deduce the order of reactivity of metals from given information;
- ☐ write balanced ionic equations;
- ☐ relate the extraction method of a metal to its position in the metal reactivity series;
- ☐ predict the feasibility of metal reactions based on the metal reactivity series.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Metals can be arranged into a list of decreasing reactivity. The list is called the reactivity series.

2 The following table summarizes the reactions of metals in the reactivity series.

<table>
<tr><th>Metal</th><th>Reaction of metal with air or oxygen</th><th>Reaction of metal with cold water or steam</th><th>Reaction of metal with dilute acids (dilute HCl and H_2SO_4)</th><th>Displacement reaction</th></tr>
<tr><td>Potassium</td><td rowspan="2">burn vigorously</td><td rowspan="2">react with cold water
metal + water ⟶ metal hydroxide + hydrogen</td><td rowspan="2">react explosively</td><td rowspan="2">K, Na and Ca react with water to give off hydrogen gas</td></tr>
<tr><td>Sodium</td></tr>
<tr><td>Calcium</td><td rowspan="4">react with decreasing vigour</td><td rowspan="3">react with steam
metal + steam ⟶ metal oxide + hydrogen</td><td rowspan="3">metal + dilute hydrochloric acid ⟶ metal chloride + hydrogen
metal + dilute sulphuric acid ⟶ metal sulphate + hydrogen</td><td rowspan="6">a metal will always displace a less reactive metal from a solution of the compound of the less reactive metal</td></tr>
<tr><td>Magnesium</td></tr>
<tr><td>Aluminium</td></tr>
<tr><td>Zinc</td><td rowspan="2">a layer of oxide forms on the surface</td><td rowspan="5">no reaction</td></tr>
<tr><td>Iron</td></tr>
<tr><td>Lead</td><td rowspan="3">no reaction</td></tr>
<tr><td>Copper</td><td rowspan="2">no reaction</td></tr>
<tr><td>Mercury</td></tr>
<tr><td>Silver</td><td rowspan="3">no reaction</td><td rowspan="3">no reaction</td></tr>
<tr><td>Platinum</td></tr>
<tr><td>Gold</td></tr>
</table>

Have you mastered?

3 Balanced chemical equations for reactions of some common metals with oxygen in the air:

Potassium	$4K(s) + O_2(g) \longrightarrow 2K_2O(s)$
Sodium	$4Na(s) + O_2(g) \longrightarrow 2Na_2O(s)$
Calcium	$2Ca(s) + O_2(g) \longrightarrow 2CaO(s)$
Magnesium	$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$
Aluminium	$4Al(s) + 3O_2(g) \longrightarrow 2Al_2O_3(s)$
Zinc	$2Zn(s) + O_2(g) \longrightarrow 2ZnO(s)$
Iron	$3Fe(s) + 2O_2(g) \longrightarrow Fe_3O_4(s)$
Lead	$2Pb(s) + O_2(g) \longrightarrow 2PbO(s)$
Copper	$2Cu(s) + O_2(g) \longrightarrow 2CuO(s)$
Mercury	$2Hg(l) + O_2(g) \longrightarrow 2HgO(s)$

Have you mastered?

4 Balanced chemical equations for reactions of some common metals with cold water or steam:

Potassium	$2K(s) + 2H_2O(l) \longrightarrow 2KOH(aq) + H_2(g)$
Sodium	$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$
Calcium	$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(s) + H_2(g)$
Magnesium	$Mg(s) + H_2O(g) \longrightarrow MgO(s) + H_2(g)$
Aluminium	$2Al(s) + 3H_2O(g) \longrightarrow Al_2O_3(s) + 3H_2(g)$
Zinc	$Zn(s) + H_2O(g) \longrightarrow ZnO(s) + H_2(g)$
Iron	$3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$

5 Balanced equations for reactions of some common metals with dilute hydrochloric acid and dilute sulphuric acid:

Metal	Chemical equation	Ionic equation
Calcium	$Ca(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2(g)$	$Ca(s) + 2H^+(aq) \longrightarrow Ca^{2+}(aq) + H_2(g)$
	$Ca(s) + H_2SO_4(aq) \longrightarrow CaSO_4(s) + H_2(g)$	—
Magnesium	$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$	$Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$
	$Mg(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2(g)$	
Aluminium	$2Al(s) + 6HCl(aq) \longrightarrow 2AlCl_3(aq) + 3H_2(g)$	$2Al(s) + 6H^+(aq) \longrightarrow 2Al^{3+}(aq) + 3H_2(g)$
	$2Al(s) + 3H_2SO_4(aq) \longrightarrow Al_2(SO_4)_3(aq) + 3H_2(g)$	
Zinc	$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$	$Zn(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$
	$Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$	
Iron	$Fe(s) + 2HCl(aq) \longrightarrow FeCl_2(aq) + H_2(g)$	$Fe(s) + 2H^+(aq) \longrightarrow Fe^{2+}(aq) + H_2(g)$
	$Fe(s) + H_2SO_4(aq) \longrightarrow FeSO_4(aq) + H_2(g)$	

6 Atoms of more reactive metals lose outermost shell electrons to form cations more readily.

7 Ionic equations for some displacement reactions:

Metal	Reaction with solution of	Ionic equation
Magnesium	aluminium compound e.g. $AlCl_3$, $Al_2(SO_4)_3$, $Al(NO_3)_3$	$3Mg(s) + 2Al^{3+}(aq) \longrightarrow 3Mg^{2+}(aq) + 2Al(s)$
	zinc compound e.g. $ZnCl_2$, $ZnSO_4$, $Zn(NO_3)_2$	$Mg(s) + Zn^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Zn(s)$
	iron(II) compound e.g. $FeCl_2$, $FeSO_4$, $Fe(NO_3)_2$	$Mg(s) + Fe^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Fe(s)$
	copper(II) compound e.g. $CuCl_2$, $CuSO_4$, $Cu(NO_3)_2$	$Mg(s) + Cu^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Cu(s)$
	silver compound e.g. $AgNO_3$	$Mg(s) + 2Ag^+(aq) \longrightarrow Mg^{2+}(aq) + 2Ag(s)$

Continued on next page

Metal	Reaction with solution of	Ionic equation
Zinc	iron(II) compound	$Zn(s) + Fe^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Fe(s)$
	copper(II) compound	$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$
	silver compound	$Zn(s) + 2Ag^{+}(aq) \longrightarrow Zn^{2+}(aq) + 2Ag(s)$
Iron	copper(II) compound	$Fe(s) + Cu^{2+}(aq) \longrightarrow Fe^{2+}(aq) + Cu(s)$
	silver compound	$Fe(s) + 2Ag^{+}(aq) \longrightarrow Fe^{2+}(aq) + 2Ag(s)$
Copper	silver compound	$Cu(s) + 2Ag^{+}(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$

8 The lower the position of a metal in the reactivity series, the more easily it can be extracted from its ores.

9 a) Compounds of a less reactive metal are more unstable. Thus, it is easier to reduce the oxide of the metal.

b) Silver oxide is reduced to silver by heating alone. Lead(II) oxide is reduced to lead by heating with carbon.

$$2Ag_2O(s) \xrightarrow{\text{heat}} 4Ag(s) + O_2(g)$$

$$2PbO(s) + C(s) \xrightarrow{\text{heat}} 2Pb(s) + CO_2(g)$$

10 When a mixture of iron(III) oxide powder and aluminium powder is ignited, a very vigorous reaction occurs — the thermit reaction.

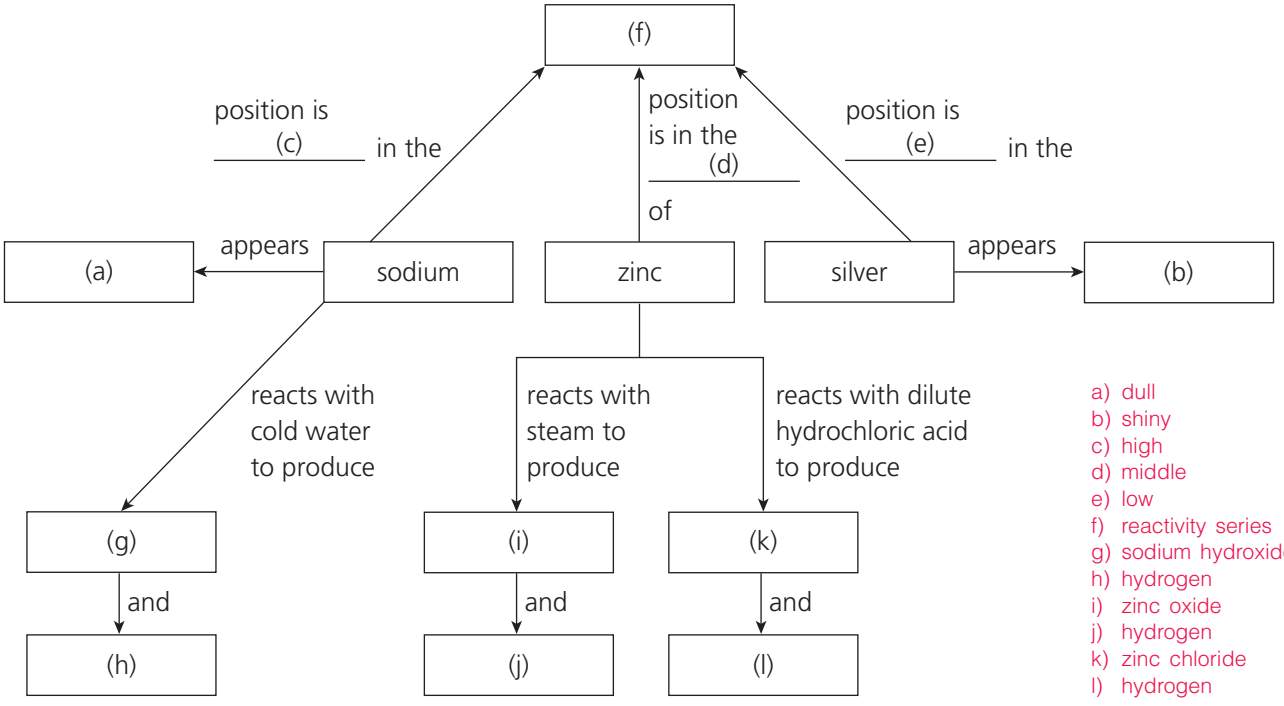
$$Fe_2O_3(s) + 2Al(s) \xrightarrow{\text{ignite}} 2Fe(s) + Al_2O_3(s)$$

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) dull
b) shiny
c) high
d) middle
e) low
f) reactivity series
g) sodium hydroxide
h) hydrogen
i) zinc oxide
j) hydrogen
k) zinc chloride
l) hydrogen

2 Balance each of the equations.

a) $4Fe + \underline{3}\ O_2 \longrightarrow \underline{2}\ Fe_2O_3$

b) $4NH_3 + \underline{5}\ O_2 \longrightarrow \underline{4}\ NO + \underline{6}\ H_2O$

c) $\underline{2}\ AgNO_3 \longrightarrow 2Ag + \underline{2}\ NO_2 + O_2$

d) $MnO_2 + \underline{4}\ HCl \longrightarrow MnCl_2 + Cl_2 + \underline{2}\ H_2O$

(Edexcel GCE O Level, Chemistry, Paper 1, Jan. 2009, 12(a)–(d))

3 We often need to show the state symbols, such as (aq), in chemical equations.

Fill in the boxes to show the state symbol for each of these chemicals at room temperature.

Chemical	Melting point (°C)	Boiling point (°C)	State symbol
A	114	184	s
B	42	713	s
C	–7	58	l
D	–101	–34	g

(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit 1, Jun. 2008, 4(a))

4 Transcribe the following word equations into chemical equations.

a) lead + oxygen ⟶ lead(II) oxide
$2Pb(s) + O_2(g) \longrightarrow 2PbO(s)$
b) potassium + water
⟶ potassium hydroxide solution + hydrogen
$2K(s) + 2H_2O(l) \longrightarrow 2KOH(aq) + H_2(g)$
c) magnesium + steam
⟶ magnesium oxide + hydrogen
$Mg(s) + H_2O(g) \longrightarrow MgO(s) + H_2(g)$
d) iron + dilute sulphuric acid
⟶ iron(II) sulphate solution + hydrogen
$Fe(s) + H_2SO_4(aq) \longrightarrow FeSO_4(aq) + H_2(g)$
e) copper + silver nitrate solution
⟶ copper(II) nitrate solution + silver
$Cu(s) + 2AgNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2Ag(s)$
f) zinc sulphide + oxygen
⟶ zinc oxide + sulphur dioxide
$2ZnS(s) + 3O_2(g) \longrightarrow 2ZnO(s) + 2SO_2(g)$

5 Write ionic equations for the following reactions.

a) magnesium + dilute sulphuric acid
$Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$
b) aluminium + dilute hydrochloric acid
$2Al(s) + 6H^+(aq) \longrightarrow 2Al^{3+}(aq) + 3H_2(g)$

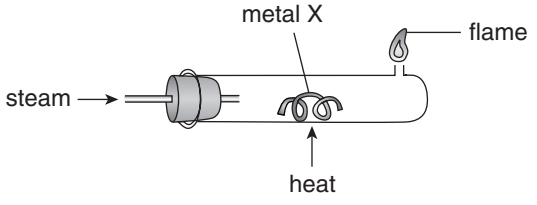
Part II Multiple choice questions

7 Which of these metals will give a lilac flame colour?

A Sodium
B Calcium
(C) Potassium
D Magnesium

(Edexcel GCE, Chemistry, Unit 2, Jan. 2009, 3)

8 Consider the following experiment.



During the experiment, a gas was liberated. The gas could burn at the end of the tube. X was probably

A copper.
B lead.
(C) magnesium.
D platinum.

Magnesium reacts readily with steam to give magnesium oxide and hydrogen.
magnesium + steam ⟶ magnesium oxide + hydrogen
Copper, lead and platinum do not react with steam.

6 a) i) heating the ore in air
ii) roasting the ore in air to obtain the oxide, then heating the oxide with carbon
iii) electrolysis of molten ore

c) zinc + iron(II) sulphate solution
$Zn(s) + Fe^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Fe(s)$
d) iron + silver nitrate solution
$Fe(s) + 2Ag^+(aq) \longrightarrow Fe^{2+}(aq) + 2Ag(s)$

6 a) Complete the following table on metal extraction methods.

Metal	Ore	Main metallic compound in the ore	Extraction method
Copper	copper pyrite	copper(II) iron(II) sulphide	(i)
Zinc	zinc blende	zinc sulphide	(ii)
Magnesium	magnesite	magnesium carbonate	(iii)

b) How does the method used to obtain a metal from its ore relate to the position of the metal in the reactivity series?

b) Metals at the top of the reactivity series are extracted by electrolysis.
Metals in the middle are extracted by reduction of their oxides with carbon.
Metals near to the bottom are extracted from their ores by heating in air.

9 X, Y and Z are metals. The table below shows the observations when each of them is put into copper(II) sulphate solution:

Metal	Observation
X	no observable change
Y	brown solid formed and colourless gas evolved
Z	brown solid formed

Which of the following arrangements correctly represents the ascending order of reactivity of the metals?

(A) X < Z < Y
B Y < Z < X
C Z < X < Y
D X < Y < Z

(HKCEE, Paper 2, 2007, 7)

10 Which of the following methods can be used to obtain magnesium from magnesium compounds?

(A) Electrolysis of a molten magnesium compound
B Electrolysis of an aqueous solution of a magnesium compound
C Heating magnesium oxide with carbon
D Heating magnesium oxide strongly

(HKDSE, Paper 1A, 2013, 5)

11 X, Y and Z are three different metals. When these metals are placed separately into an aqueous solution of tin(II) nitrate, a spongy layer of tin is formed only on X. When each of the oxides of these metals is heated strongly, only the oxide of Y gives a metallic lustre. Which of the following represents the arrangement of these metals in decreasing order of reactivity?

A X > Y > Z
(B) X > Z > Y
C Y > X > Z
D Z > X > Y

(HKDSE, Practice paper 1A, 2012, 6)

12 The table below shows whether displacement reaction occurs between metals W, X, Y and Z with their ions.

	W	**X**	**Y**	**Z**
$W^{2+}(aq)$		✘	✔	✔
$X^{+}(aq)$	✔		✔	✔
$Y^{2+}(aq)$	✘	✘		✔
$Z^{+}(aq)$	✘	✘	✘	

'✔' represents displacement reaction occurs
'✘' represents displacement reaction does not occur

Which of the following represents the arrangement of these metals in increasing order of reactivity?

(A) X < W < Y < Z
B X < Z < W < Y
C Z < X < Y < W
D Z < Y < W < X

Metal Z can displace the other three metals from solutions of their ions. Thus, Z is the most reactive.

Metal X cannot displace any of the metals from solutions of their ions. Thus, X is the least reactive.

13 The following equation represents the reaction between an oxide of lead and hydrogen:

$$w Pb_3O_4(s) + x H_2(g) \longrightarrow y Pb(s) + z H_2O(l)$$

What are the values of *w*, *x*, *y* and *z* respectively in the equation?

	w	x	y	z
(A)	1	4	3	4
B	2	4	2	3
C	3	3	4	2
D	4	2	3	4

Directions: Questions 14–15 refer to the following information:

Iron and copper can be extracted from their oxides by reduction with carbon. Aluminium cannot be extracted from its oxide in this way.

Copper can be found in the Earth's crust as the metal itself. Iron and aluminium are found only as compounds in the Earth's crust.

14 Which row in the table shows the order of reactivity of the four elements in the passage above?

	Most reactive → Least reactive			
A	copper	iron	aluminium	carbon
(B)	aluminium	carbon	iron	copper
C	carbon	copper	iron	aluminium
D	iron	aluminium	copper	carbon

(AQA GCSE (Higher Tier), Chemistry, Unit C1a, Jun. 2011, 5(a))

Iron and copper can be extracted from their oxides by reduction with carbon while aluminium cannot. Thus, aluminium is more reactive than iron and copper.

15 Iron can be extracted from iron(III) oxide, Fe_2O_3, by mixing the oxide with carbon and heating strongly in air. Which row in the table below shows the name of the process and the products of the reaction?

	Name of the process	Products of the reaction
A	electrolysis	Fe, O_2 and CO_2
(B)	reduction	Fe and CO_2
C	electrolysis	Fe and CO_2
D	reduction	Fe, O_2 and CO_2

(AQA GCSE (Higher Tier), Chemistry, Unit C1a, Jun. 2011, 5(b))

iron(III) oxide + carbon $\xrightarrow{\text{heat}}$ iron + carbon dioxide

16 A scientist investigated some reactions of three metals, K, L and M.

- Metal L can be extracted by mixing its oxide with carbon and heating.
- Metal M CANNOT be extracted by mixing its oxide with carbon and heating.
- Hydrogen will displace metal K from its oxide but will NOT displace metal L from its oxide.
- The oxide of metal K is decomposed to the metal on heating alone.

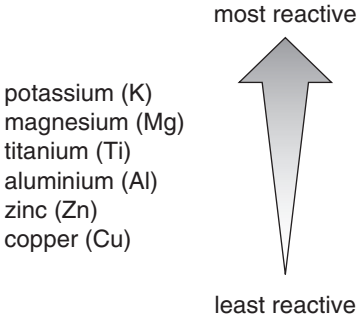
The order of reactivity for the three metals, with the most reactive first, is

A K, L, M.
(B) M, L, K.
C L, K, M.
D K, M, L.

L can be extracted by carbon reduction while M cannot. Thus, M is more reactive than L.
Hydrogen can displace K from its oxide but not L. Thus, L is more reactive than K.
∴ M is the most reactive metal and K is the least reactive metal.

(AQA GCSE (Higher Tier), Chemistry, Unit C1a, Nov. 2012, 9(a))

17 The order of reactivity for some metals is shown below.



Which of the following shows a correctly balanced equation for a possible reaction?

A $ZnO + C \longrightarrow Zn + CO_2$
B $Cu + MgO \longrightarrow CuO + Mg$
C $2Mg + K_2O \longrightarrow 2MgO + 2K$
(D) $2Al + 3ZnO \longrightarrow 3Zn + Al_2O_3$

(AQA GCSE (Higher Tier), Chemistry, Unit C1a, Nov. 2012, 9(b))

Option B — Cu is less reactive than Mg. There is NO reaction between Cu and MgO.

Option C — Mg is less reactive than K. There is NO reaction between Mg and K_2O.

18 (2) Magnesium can displace lead from lead(II) nitrate solution.

$Mg(s) + Pb(NO_3)_2(aq) \longrightarrow Mg(NO_3)_2(aq) + Pb(s)$

(3) Heating lead(II) oxide with carbon powder produces lead.

$2PbO(s) + C(s) \xrightarrow{\text{heat}} 2Pb(s) + CO_2(g)$

18 From which of the following processes can lead be obtained in a school laboratory?

(1) Heating lead(II) oxide strongly
(2) Adding magnesium to lead(II) nitrate solution
(3) Heating lead(II) oxide with carbon powder

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

19 Rubidium is a Group I element below potassium in the periodic table. Which of the following statements about rubidium are correct?

(1) Rubidium loses electrons more readily than potassium does.
(2) Rubidium can be extracted by electrolysis of its molten ore.
(3) Rubidium was discovered before potassium.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(1) Rubidium is more reactive than potassium.

20 Which of the following processes would produce hydrogen gas?

(1) Adding calcium to cold water
(2) Adding iron to dilute hydrochloric acid
(3) Passing steam over zinc

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

(1) Calcium reacts with water to give calcium hydroxide and hydrogen.

$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(s) + H_2(g)$

(2) Iron reacts with dilute hydrochloric acid to give iron(II) chloride and hydrogen.

$Fe(s) + 2HCl(aq) \longrightarrow FeCl_2(aq) + H_2(g)$

(3) Zinc reacts with steam to give zinc oxide and hydrogen.

$Zn(s) + H_2O(g) \longrightarrow ZnO(s) + H_2(g)$

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

21 Write balanced equations for the following reactions.

a) magnesium + oxygen ⟶ magnesium oxide
$2Mg + O_2 \longrightarrow 2MgO$

b) potassium hydroxide + carbon dioxide ⟶ potassium carbonate + water
$2KOH + CO_2 \longrightarrow K_2CO_3 + H_2O$

c) aluminium sulphate + sodium hydroxide ⟶ aluminium hydroxide + sodium sulphate
$Al_2(SO_4)_3 + 6NaOH \longrightarrow 2Al(OH)_3 + 3Na_2SO_4$

(Edexcel GCE O Level, Chemistry, Paper 1, May 2010, 6(a)–(c))

22 For each of the following experiments, state ONE observable change and write a chemical equation for the reaction involved.

Refer to page T32.

a) Zinc was put into dilute hydrochloric acid.

b) Sodium was heated in a Bunsen flame.

c) Lead(II) oxide was heated with carbon powder.

d) Copper was put into silver nitrate solution.

e) Copper(II) oxide was heated with magnesium.

23 Electric wires are usually made of copper because copper is a good electrical conductor.

a) How is electricity conducted through copper?

Put a tick '✔' in the box next to the BEST answer.

Anions move to the anode, cations to the cathode.	☐
Positive ions move through the lattice.	☐
Electrons move between ions in the lattice.	✔
Electrons move between atoms in the lattice.	☐
Atoms vibrate within the lattice.	☐

b) Copper is extracted from its ore using a blast furnace.

The blast furnace melts all the reactants and allows them to react.

One reaction that takes place is that of molten copper(I) sulphide with molten copper(I) oxide to produce molten copper and sulphur dioxide gas.

Complete the reaction by putting state symbols into the brackets.

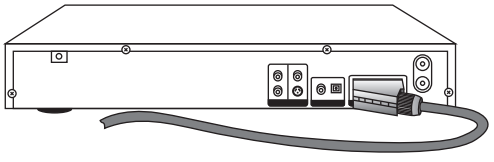
Cu_2S [...l...] + $2Cu_2O$ [...l...] ⟶ $6Cu$ [...l...] + SO_2 [...g...]

c) Another of the reactions that takes place is

$2Cu_2S$ + [3] O_2 ⟶ [2] Cu_2O + [2] SO_2

Complete the boxes to balance the equation.

d) The cable connecting a TV to a DVD player sometimes has gold plated contacts.



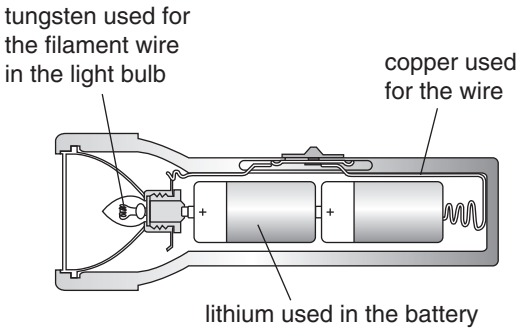
There are two different reasons why the gold is used for contacts.

Put a tick '✔' in the boxes next to the TWO best reasons why gold is used for contacts.

Gold bends easily.	☐
Gold is a very unreactive metal.	✔
Gold has the same colour as copper.	☐
Gold is a rare metal.	☐
Gold can be plated very thinly.	☐
The nucleus of a gold atom holds its outer electrons quite weakly.	✔

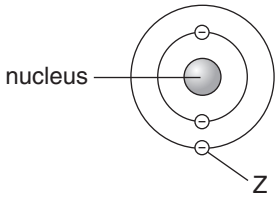
(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit 2, Jan. 2009, 3)

24 The diagram shows a circuit that is used in a torch. Electrons flow through this circuit.



a) Why is copper used for the wire?
Copper can conduct electricity.

b) The diagram shows the structure of an atom of lithium.



Name the particle labelled Z. Electron

c) The table shows some properties of the metals used in the electric circuit.

Metal	Melting point (°C)	Boiling point (°C)	Reaction with oxygen
Copper	1 083	2 582	Reacts slowly to form a thin oxide layer on surface
Lithium	179	1 317	Reacts rapidly to form oxide
Tungsten	3 370	5 930	Reacts only when very hot to form oxide

i) Use information from the table to suggest the order of reactivity for copper, lithium and tungsten.
Lithium > copper > tungsten

ii) The filament wire glows because it gets very hot.

Use information from the table to suggest ONE reason why tungsten is used for the filament wire in the light bulb. It has a very high melting point. / It can withstand the highest temperature.

d) The gas used in the light bulb is argon.

Circle the correct word to complete the sentence.

Argon is used in the light bulb because it is dense. / solid. / unreactive.

(AQA GCSE (Foundation Tier), Chemistry, Unit 1, Jun. 2009, 1)

25 Calcium and sodium are both reactive metals.

A small piece of each metal is added to separate troughs of water.

The metals react with water as shown in these equations:

calcium + water ⟶ compound A + gas X

sodium + water ⟶ compound B + gas X

a) i) State ONE observation that would be the same during both reactions.
Bubbles evolve / fizzing

ii) State ONE observation that could be made during the reaction between sodium and water, but not during the reaction between calcium and water.
Sodium moves rapidly on the water surface.

b) i) What is the name of compound A?
Calcium hydroxide

ii) What is the formula of compound B?
NaOH

c) Identify gas X and describe a test and the expected result for this gas.
Hydrogen. It gives a 'pop' sound with a burning splint.

d) What colour does compound B give in a flame test? Golden yellow

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2, Jun. 2010, 1(a)–(d))

26 The following metals are arranged in the order of decreasing reactivity.

metal *a* > metal *b* > metal *c* > iron > metal *d* > metal *e* > metal *f*

Based on the information given above, answer the following questions.

a) Which metal is most likely to tarnish rapidly in air? Explain your answer.

b) Which metal is most likely to be found free in nature? Explain your answer.

c) Would you expect metal *d* to react with cold water? Explain your answer.

d) Suggest how metal *a* can be extracted from its ore.

e) What would be formed when the oxide of metal *d* is heated with metal *b*? Explain your answer.

f) Suggest an experiment to show that metal *b* is more reactive than iron.

g) Can we store a solution of the nitrate of metal *c* in a container made of metal *d*? Explain your answer.

27 Carbon can be used to extract metals from certain metal oxides.

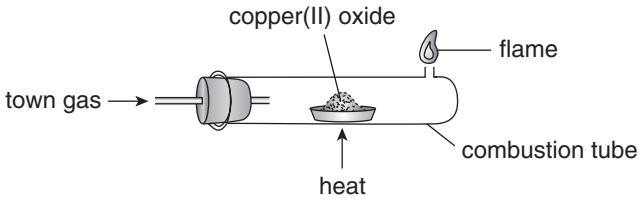
a) Suggest how copper can be extracted from copper(II) oxide using carbon. State the expected observation.

b) Explain whether carbon can also be used to extract magnesium from magnesium oxide.

(HKCEE, Paper 1, 2009, 2(b))

26 a) Metal *a*
Because it is the most reactive metal and thus reacts rapidly with oxygen in air to form an oxide layer on the metal surface.
b) Metal *f*
Because it is the least reactive metal and thus does not form stable compounds readily.
c) No.
Iron has no reaction with cold water and metal *d* is less reactive than iron.
d) Electrolysis of its molten ore
e) Metal *d* is formed. Metal *b* is more reactive than metal *d*. It can remove oxygen from the oxide of *d*.
f) Add metal *b* to iron(II) sulphate solution. Metal *b* can displace iron from iron(II) sulphate solution.
g) Yes.
Since metal *d* is less reactive than metal *c*, no displacement reaction occurs.

28 The following experimental set-up was used for converting copper(II) oxide to copper.



The hydrogen gas and carbon monoxide gas in town gas converted the copper(II) oxide to copper.

a) State ONE observable change you expect in the experiment. The black copper(II) oxide changed to reddish brown copper.

b) Suggest ONE way to show that a metal was formed in this experiment.
The metal obtained can conduct electricity.

c) Name the type of reaction that occurs when copper(II) oxide was heated with town gas.
Reduction

d) Write a chemical equation for the reaction between

i) copper(II) oxide and hydrogen; and
$CuO(s) + H_2(g) \longrightarrow Cu(s) + H_2O(l)$

ii) copper(II) oxide and carbon monoxide.
$CuO(s) + CO(g) \longrightarrow Cu(s) + CO_2(g)$

e) Suggest why it was necessary to burn the residual gas in the set-up.
Town gas is flammable.

f) At the end of the reaction, heating was stopped. However, town gas was still passing through the combustion tube while it was cooling down. Suggest a reason for this.
To prevent the re-oxidation of the hot copper.

29 Copper forms when magnesium reacts with copper(II) nitrate solution. The ionic equation for the reaction is

$Cu^{2+}(aq) + Mg(s) \longrightarrow Cu(s) + Mg^{2+}(aq)$

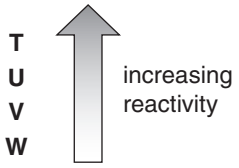
a) What does this reaction indicate about the reactivity of copper?
Less reactive than magnesium

b) Describe the colour change of the solution if an excess of magnesium is added.

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2, Jun. 2009, 5(d))

The colour of the solution changes from blue to colourless.

Q30 The order of reactivity of these metals is



a) When metal T is added to a solution of the sulphate of metal U, a displacement reaction occurs.

Ionic half-equations for the reactions occurring are:

$$T(s) \longrightarrow T^{2+}(aq) + 2e^-$$

and

$$U^{2+}(aq) + 2e^- \longrightarrow U(s)$$

Suggest ionic half-equations for the reactions occurring when metal V is added to a solution of the sulphate of metal W. $V(s) \longrightarrow V^{2+}(aq) + 2e^-$ $W^{2+}(aq) + 2e^- \longrightarrow W(s)$

b) The overall ionic equation for the reaction that occurs when metal T is added to a solution of the sulphate of metal U is

$$T(s) + U^{2+}(aq) \longrightarrow T^{2+}(aq) + U(s)$$

Suggest an overall ionic equation for the reaction that occurs when each of the following are mixed.

If no reaction occurs, write no reaction.

i) Metal V is added to a solution of the sulphate of metal U. No reaction

ii) Metal T is added to a solution of the sulphate of metal W. $T(s) + W^{2+}(aq) \longrightarrow T^{2+}(aq) + W(s)$

(Edexcel IGCSE (Foundation and Higher Tiers), Chemistry, Paper 3, Nov. 2010, 5(e)–(f))

Q31 The table below lists some information about three metals, X, Y and Z.

Metal	X	Y	Z
Atomic number	20	—	—
Reaction with cold water	a colourless gas evolves	no observable change	no observable change
Heating metal oxide	no observable change	a solid with metallic lustre forms	no observable change

a) Name metal X.
Calcium

b) X reacts with cold water to give a colourless gas.

i) Name the colourless gas and write a chemical equation for the reaction involved. Hydrogen
$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(aq) + H_2(g)$

ii) In a practical lesson, a student added a few granules of X into a beaker of cold water.

Draw a labelled diagram to show how the student could collect the gas produced.

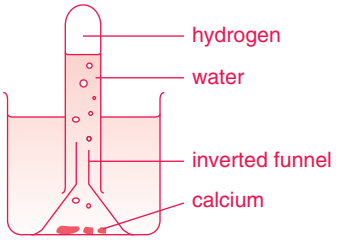
c) Arrange the three metals in order of increasing reactivity. Explain your answer.

d) The following table lists the year of discovery of the three metals.

Metal	Year of discovery
X	1808
Y	ancient
Z	1746

Suggest a relationship between the reactivity and the year of discovery of the metals.

31 b) ii)



31 c) Reactivity: Y < Z < X

Y is the least reactive as it can be extracted from its oxide by heating alone but the other two metals cannot.

X is more reactive than Z as it reacts with cold water but Z does not.

d) Less reactive metals were discovered first and more reactive metals were discovered later.

32 Titanium is used for replacement hip joints because it has a low density, is strong and does not corrode.

Titanium is extracted from titanium dioxide (TiO_2) in three stages.

a) *Stage 1*

Titanium dioxide is converted into titanium(IV) chloride ($TiCl_4$) because the metal cannot be extracted from its oxide by reduction with carbon.

i) What does 'reduction' mean?
Removal of oxygen

ii) Balance the chemical equation for the conversion of titanium dioxide to titanium(IV) chloride.

$$_1_\ TiO_2(s) + _2_\ Cl_2(g) + _1_\ C(s) \longrightarrow _1_\ TiCl_4(l) + _1_\ CO_2(g)$$

iii) Chemical equations are always balanced. Explain why.
No atoms are lost / made during a chemical reaction.

b) *Stage 2*

Titanium is extracted from the titanium(IV) chloride by reacting it with sodium at 1 000 °C in a reactor.

The only other substance in the reactor is argon gas.

$$TiCl_4(l) + 4Na(l) \longrightarrow Ti(l) + 4NaCl(l)$$

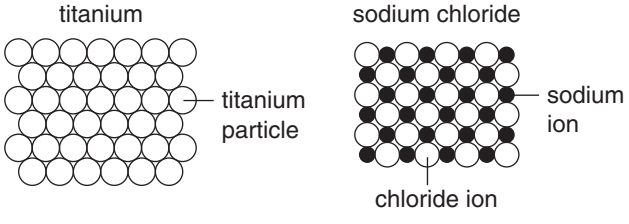
i) What does this tell you about the reactivity of sodium when compared with titanium?
Sodium is more reactive than titanium.

ii) Suggest why the reactor contains argon and NOT air.
Sodium / titanium would react with oxygen / air.

c) *Stage 3*

After *Stage 2* the titanium is separated from the products by washing off the sodium chloride with water.

The diagrams show sections through the lattice of titanium metal and the lattice of sodium chloride.



How do the diagrams show that:

i) titanium is an element?
All the atoms are the same.

ii) sodium chloride is a compound?
Different types of ions bonded together.

(AQA GCSE (Higher Tier), Chemistry, Unit 1, Jan. 2011, 4)

33 The following article appeared recently in the Manchester Gazette.

Sodium Drum Blaze Scare

A 20 litre drum containing sodium burst into flames when it reacted violently with rainwater at a Manchester factory. It is believed that the sodium, which is normally stored under oil, had been accidentally left outside with the lid off.

A factory worker put out the blaze before the fire services arrived, and a leading fire fighter said, 'It was fortunate that potassium wasn't involved as it would have reacted more violently and exploded. These Group 1 alkali metals can be very dangerous'.

a) Group 1 metals are stored under oil.

Suggest why.
To prevent them from reacting with the air.

b) Balance the equation which represents the reaction between sodium and water.

$$_2_\ Na(s) + _2_\ H_2O(l) \longrightarrow _2_\ NaOH(aq) + _1_\ H_2(g)$$

c) Explain why the Group 1 metals are called the alkali metals. These metals react with water producing an alkaline solution.

d) Explain, in terms of electrons, why potassium reacts more violently than sodium.

(AQA GCSE (Higher Tier), Chemistry, Unit 3, June 2008, 4)

A potassium atom is larger than a sodium atom.

There is less attraction between the outermost shell electron and the nucleus in a potassium atom.

A potassium atom loses its outermost shell electron more readily than a sodium atom.